

FaunaCodec: ROI-Aware Video Compression and Reconstruction for Wildlife Monitoring

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Abstract—Remote, stationary wildlife-monitoring cameras (camera traps) must transmit video over low-bandwidth uplinks from power-constrained edge devices, yet the animal of interest occupies only a fraction of an otherwise static frame while standard codecs compress it uniformly. We introduce FaunaCodec, a modular region-of-interest (ROI)-aware compression framework that allocates quality unequally between subject and background: an off-the-shelf detector localizes the animal, and two independently scheduled streams encode the subject at high quality and the background aggressively. On the server, video frame interpolation recovers the dropped frames and a replaceable diffusion-based super-resolver restores delivery resolution. Detection, compression, and upscaling are independently replaceable modules, so the same architecture drives both a learned neural codec and conventional codecs. On field-collected wildlife footage, FaunaCodec cuts transmitted data by roughly 46–80× (corpus mean, with wide per-clip variance) relative to the camera’s native recording while keeping the subject detectable. Full-reference and no-reference quality metrics rank the generative super-resolvers in opposite orders, so we treat upscaled output as a perceptual aid rather than photometric evidence. The saving traces to background bit-allocation rather than the two-stream structure, which we position as a codec-agnostic route to ROI quality for codecs without a native ROI mode.

Index Terms—camera traps, wildlife monitoring, region-of-interest video coding, neural video codec, edge AI, video frame interpolation, diffusion-based super-resolution, no-reference quality assessment

I. INTRODUCTION

Remote wildlife monitoring depends on camera traps that record animal activity in habitats far from power and network infrastructure [1], [2], running continuously and unattended while the animal of interest occupies only a fraction of each otherwise static frame. Sustained deployment is bottlenecked by storage (uncompressed $1920 \times 1080/30$ fps video generates hundreds of GB per hour, and even with standard codecs [3], [4], [5] continuous recording fills storage within hours), connectivity (remote uplinks run tens to hundreds of kilobits per second [6]), and hardware (battery- or solar-powered cameras prioritize low power over sensor quality and mostly capture at 720p or 1080p). A codec that allocates bits selectively to the region of interest (ROI) [7] and server-side upscaling can address all three, but their integration into a deployable framework remains open.

We present FaunaCodec, a modular framework built around ROI-aware dual-stream compression: an off-the-shelf detector

localizes the subject, two independent codec streams encode the ROI at high quality and the background aggressively, packaged with the detection metadata into one archive; the server composites the streams, interpolates dropped frames, and applies a replaceable upscaling stage. Our contributions:

- A codec-agnostic ROI-aware split, giving region-adaptive quality even to a learned codec (DCVC) with a single global quality parameter; its separate streams could further support unequal error protection under lossy transport.
- A motion-aware frame-selection policy sampling the two streams on independent schedules, with dropped frames recovered by server-side interpolation.
- A controlled characterization against an in-stream ROI baseline at matched fidelity, with a detector-agreement proxy and a mechanism ablation, locating the compression gain in background bit-allocation rather than the split itself.
- An empirical evaluation across codec and upscaler choices on field-collected wildlife footage.

Code and model weights, and the evaluation corpus are publicly available.¹

II. RELATED WORK

a) Video Coding and ROI: Block-based hybrid codecs, H.264/AVC [3], H.265/HEVC [4], AV1 [5], remain the deployed baseline, allocating quality uniformly with no notion of semantic value; learned codecs such as DCVC-FM [8], our default backend, match or exceed them but are equally subject-unaware. Allocating bits non-uniformly toward task-relevant regions is long-standing; end-to-end ROI neural video compression spatially scales the latent to concentrate rate [7], keeping bit allocation inside a single bitstream tied to one learned codec. We instead split subject and background into two fully independent streams at separate quality points, a codec-independent structural choice rather than a rate-distortion one (Section III-A2). Video coding for machines [9] and learned image coding for machines [10] optimize compression for a downstream analytic rather than a human viewer; we keep the codec fixed and evaluate with a detector-agreement proxy instead (Section IV-C). Dropping temporally redundant frames

¹<https://github.com/imics-lab/FaunaCodec>

and restoring the frame rate at the server is classical frame-rate up-conversion; we adopt the learned all-pairs multi-field interpolator AMT [11] to synthesize intermediate frames from sparse anchors (Section III-B1).

b) Generative Reconstruction and Wildlife Applications: Single-step distilled diffusion super-resolvers such as OSediff [12] make diffusion-based upscaling practical; we treat the upscaler as an interchangeable, evaluated backend. Camera traps produce large volumes of empty or redundant footage, motivating automated triage: MegaDetector [1] and deep classifiers [2] address detection and species recognition, and platforms such as Wildlife Insights [6] address storage and sharing at scale; these analyze footage after capture, whereas we use detection at capture to drive compression.

III. METHOD

Figure 1 gives an end-to-end overview of the framework.

A. Edge Node

The edge node, a solar-powered device deployed unattended in the field, runs ROI detection and dual-stream compression on-device so only the compact archive traverses the low-bandwidth uplink.

1) ROI Detection: The animal is localized as an ROI the compression stage treats preferentially, by an off-the-shelf, swappable detector, MegaDetector V6 (MDV6-yolov9-c) [13] by default. Keyframe detections, spaced $K=15$ frames apart, are propagated across intervening frames by sparse optical-flow tracking, emitting ROI boxes with track identifiers, class labels, and confidence scores for every frame.

2) Dual-Stream Compression:

a) Dual-stream DCVC encoding: DCVC-FM [8], hereafter DCVC, is a learned neural video codec (intra model DMCI, inter model DMC) parameterized by a quantization parameter $QP \in \{0, \dots, 63\}$, higher is better. We encode ROI and background as independent bitstreams at deliberately mismatched quality, $QP_{ROI}=63$ and $QP_{BG}=25$, both encoding the full frame, the high-quality ROI composited over the low-quality background at the decoder (Section III-B2); the first frame of each stream is DMCI, the rest DMC with a feature-adaptor reset every 32 frames. The construction is codec-agnostic, carrying over to any backend by replacing DCVC's QP with that codec's own quality control (Section IV-B); block codecs additionally mask the ROI stream to the detection box, which the neural path does not.

b) Adaptive frame selection: A motion-aware policy selects frames independently on the two streams: the ROI stream maintains a hysteresis-gated STILL/MOTION state from pixel-difference and centroid-displacement evidence, sampling densely during motion and sparsely during stillness, while the background stream follows an independent schedule, sampled tightly while a detection is present and widely otherwise.

3) Archive File Formation: The compressed unit is a single ZIP archive containing the two encoded bitstreams, per-frame bounding boxes with track identifiers, frame-selection metadata, and codec metadata; this archive is the unit of transmission.

B. Server Node

The server node reverses the edge-side encoding before optional upscaling: the archive's metadata fixes the decoder configuration and places each decoded frame back on the original timeline.

1) Decompression and AMT Reconstruction of Dropped Frames: The server decodes both bitstreams (DMCI for I-frames, DMC for P-frames), giving decoded ROI and background frames $\mathbf{F}_t^{ROI}$ and $\mathbf{F}_t^{BG}$ at time t . Because the streams are sampled on independent, sparse timelines, an output index may be missing its ROI frame, background frame, or both. The background at a dropped index is recovered by linear interpolation between its neighboring anchors; the ROI, carrying the moving animal, uses AMT [11], an all-pairs multi-field flow interpolator, cropped to the union box of consecutive anchors and restricted to the subject. Each synthesized crop is pasted onto the interpolated background and merged through the soft-mask compositing of Section III-B2; when interpolation is disabled the crops blend linearly, degrading gracefully to a classical baseline. Residual flicker where the two streams meet is suppressed by a motion-aware exponential moving average over the ROI.

2) Video Frame Reassembly: A binary mask, filled from stored bounding boxes and feathered by a distance transform into a soft alpha channel $\tilde{\mathbf{M}}_t$ that suppresses visible seams, blends the two decoded streams into the composite frame $\mathbf{F}_t = \mathbf{F}_t^{ROI} \odot \tilde{\mathbf{M}}_t + \mathbf{F}_t^{BG} \odot (1 - \tilde{\mathbf{M}}_t)$.

3) Upscaler Module: The upscaler (Fig. 1) is the final, *replaceable* stage: it takes composited frames $\mathbf{F}_t$ at decoded resolution and emits them at a target resolution ($2\times$ by default), one backend active per run. We group backends by how strongly they synthesize detail, trading full-reference fidelity for perceptual sharpness: Lanczos-windowed bicubic (no added detail, the fidelity reference), S3Diff [14] (degradation-guided one-step diffusion, moderate synthesis), OSediff [12] (distilled one-step diffusion, fast enough for delivery, plausible but ground-truth-independent texture), and PiD [15] (pixel-space diffusion whose checkpoint is a native $4\times$ model, so we downscale its input accordingly, yielding the densest hallucinated texture); the framework also supports video-native models that consume a temporal window at substantially higher compute cost. A per-channel AdaIN color fix corrects global tint; the active backend's output is the delivered frame $\mathbf{S}_t$.

IV. EXPERIMENTS

A. Dataset

Footage comes from a field-deployed edge node (Jetson Nano class, battery/solar) recording only when a motion-then-detector cascade confirms a target species, so every clip is event-centric. We evaluate on a fixed 20-clip subset spanning seven animal categories (birds, cattle, deer, fox, raccoon, skunk, wild turkey) under day/night conditions, single- and multi-subject scenes; the capture-gating detector is distinct from the framework's ROI detector (Section III-A1). Reconstruction quality is scored against the original input clip, the only ground truth available in

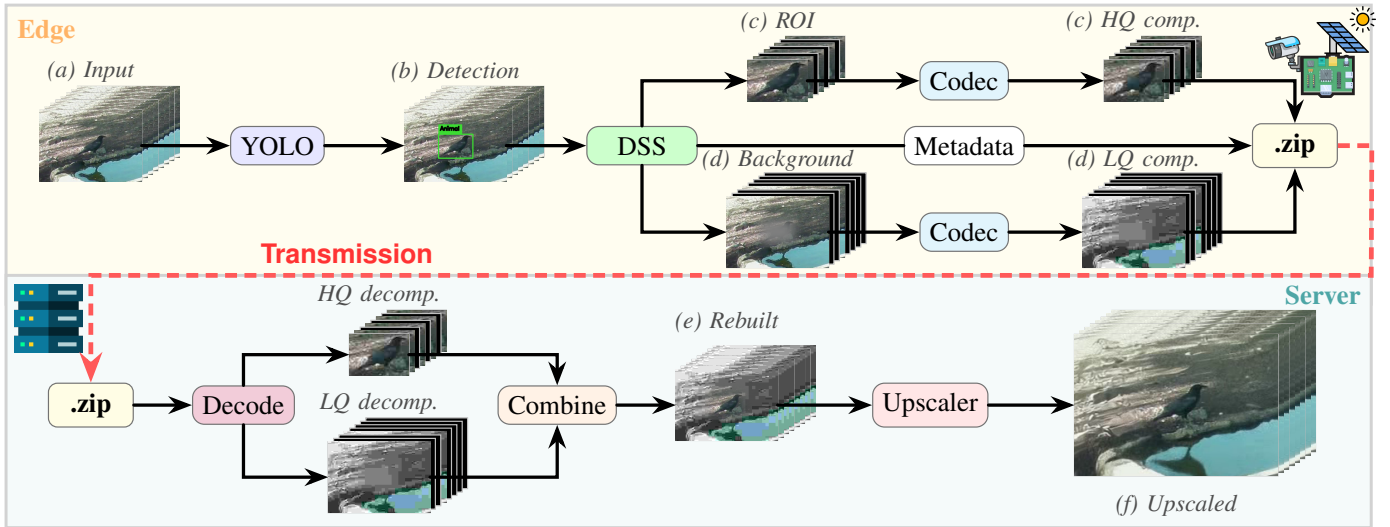


Fig. 1: FaunaCodec framework using frames from a wildlife camera trap recording. (a) An input frame is processed by an object detector to produce a bounding box. (b) The dual-stream split (DSS) divides the frame into a high-quality ROI stream (c) and a low-quality background stream (d), each encoded independently into separate codec bitstreams (H.264, H.265, AV1, or DCVC). On the server, both streams are decoded and (e) composited. The composite is then (f) upscaled by an interchangeable backend (bicubic, S3Diff, OSEDiff, PiD, Wan2.1, or CogVideoX).

the field. Compression runs inside a Docker container capped at 4 CPU cores and 4 GB RAM, GPU access only where needed; decompression and upscaling run on the less-constrained server.

B. Codec and Module Backends

Because the framework is codec- and upscaler-agnostic, we evaluate each stage by swapping backends while holding the rest fixed. On compression we compare DCVC against H.264, H.265, and AV1, with ROI and background in the same codec family per run; the QP scales run in opposite directions (lower is better for the block codecs, ROI 20 vs. background 38; higher for DCVC, ROI 63 vs. background 25, Section III-A2). We report the deterministic int16 DCVC runtime throughout: with the current reference implementation, the floating-point entropy coder is not bit-exact between the ARM edge encoder and the x86 server decoder, so float-encoded streams decode corrupted. We regard both this and the int16 runtime’s higher edge cost (Section V-A) as artifacts of the present implementation, not properties of learned coding. On upscaling, we evaluate the backends of Section III-B3.

C. Evaluation Metrics

We assess quality with PSNR, MS-SSIM [16], LPIPS [17], VMAF [18], NIQE [19], CLIP-IQA [20], and MANIQA [21]: the first four full-reference (FR), against the camera’s own lossy recording, and the last three no-reference (NR), since generative upscaling synthesizes detail no FR metric rewards fairly (Section VI-B). The corpus codec comparison reports FR metrics over both the full frame and the ROI region; the end-to-end matrix reports PSNR, VMAF, and the three NR metrics. For the controlled architecture comparison (Section V-B) we add a *detector-agreement* proxy: MegaDetector V6 runs on each reconstruction and on the original clip, whose output serves as pseudo-reference, and we report *recall* (fraction of

original detections recovered, $\text{IoU} \geq 0.5$) and *confidence preservation* (ratio of summed confidence). The proxy measures the detectability that compression and reconstruction preserve, not absolute accuracy. System cost is archive size and compression ratio, plus encode and decode runtime relative to real time.

D. Experimental Design

a) End-to-end framework: The central experiment is a matrix over codec backends and upscaling modules, run end to end from compression through upscaled output (Table II).

b) Compression layer and controlled ablation: The codec comparison runs across the 20-clip subset independent of upscaling. To isolate what the split contributes we run three controlled studies. (i) *In-stream ROI baseline:* for H.264/H.265, the split vs. in-stream encoding (`addroi`) at matched subject fidelity and background level, same ROI geometry and frame set, frame dropping off (Section VI-A); rate-distortion curves compared at equal subject PSNR via the Bjøntegaard convention [22]. (ii) *Mechanism ablation:* disable frame dropping and background quality reduction one at a time, measuring archive growth. (iii) *Downstream-task proxy:* detector agreement (Section IV-C) for both architectures at matched total bytes. The neural backend has no native in-stream mode, so (i)/(iii) run on block codecs; (ii) covers all backends.

V. RESULTS

A. Codec Comparison Across the Corpus

Table I reports per-codec means over the 20 clips: compression in the resource-capped container, ROI and background streams in the same codec family, quality after AMT reconstruction against the original. The ratio is against the camera’s native single-stream H.264/MP4 recording, the baseline a field deployment would otherwise transmit; scoring after the full decode path folds in selection- and interpolation-induced loss.

TABLE I: Codec comparison; corpus-mean values. Archive size is the transmitted artifact; compression ratio is source size over archive size. ROI (green) and full-frame (lavender) metrics; the gap reflects the deliberate quality split. Plain float DCVC is omitted (Section IV-B).

Codec	Arch.(KB)		Ratio	PSNR		MS-SSIM		VMAF		PSNR		MS-SSIM		LPIPS		VMAF	
	↓	↑		↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↓	↓	↑	↑
H.264	681.1	47.4	52.74	0.998	90.93	30.11	0.922	0.421	49.23								
H.265	487.0	79.6	52.50	0.998	90.50	29.04	0.901	0.459	40.04								
AV1	937.9	45.9	52.90	0.998	90.93	32.14	0.976	0.272	79.85								
DCVC-int16	600.9	49.6	54.10	0.998	91.67	31.40	0.930	0.409	53.32								

The ROI-aware split works: every codec holds the subject at high quality while spending far fewer bits on the background, and every backend cuts archive size sharply (45.9× AV1 to 79.6× H.265), so the bandwidth advantage holds for all four. Archive size and ratio are highly content-dependent (coefficient of variation, CV, 0.67–0.79) while perceptual and structural metrics stay stable (CV < 0.05); paired tests on compression ratio with Holm correction [23] ($n = 20$, $\alpha = 0.05$) place H.265 highest, with AV1, H.264, and DCVC-int16 tied. System cost separates a light edge encode from a heavier server decode: the block codecs encode at or faster than real time on CPU alone, while the int16 DCVC runtime in its current form needs a GPU and encodes about 6× slower, and decode stays within the server budget for every codec.

B. Where the Compression Gain Comes From

Disabling background crushing inflates the archive by +196% (DCVC-int16) to +825% (H.264); disabling frame dropping alone costs only +2.6% to +16.9% ($n = 20$), 12–200× less. The 46–80× headline is therefore a background-bit-allocation effect an in-stream ROI quantizer can also realize, not evidence for the two-stream structure. Testing this directly, at matched subject fidelity in-stream ROI encoding on the two block codecs with a native ROI mode (Section IV-D) is usually as cheap or cheaper, winning every clip under H.264 (median split penalty +11%) and 14 of 20 under H.265 (median +9%). The split leads only on five large-bounding-box daytime H.265 clips, where the ROI covers most of the frame and the distinction matters least; in-stream’s clearest wins fall on the small, localized nocturnal subjects that matter most for wildlife. Detector agreement at matched bytes shows no meaningful difference either (dual minus in-stream median recall/confidence: −0.018/−0.006 H.264, −0.006/−0.003 H.265; the split leads on 13 of 40 clip-codecs on recall and 15 of 40 on confidence), if anything favoring in-stream. The split is not more rate-efficient and gives no downstream-task edge; its case is structural (Section VI-A).

C. End-to-End Framework Matrix

Table II reports the full framework at corpus scale, pairing each codec-upscaler configuration with its reconstruction quality and upscaling cost, under a different encoding budget and scoring protocol than Table I, so the two are not comparable cell-for-cell.

TABLE II: End-to-end quality and cost by codec and upscaler; corpus means over the 12 clips whose full grid completed (all 11 multi-animal clips plus one single-animal clip). Archive size (KB) is fixed at compression time. FR (blue): (1) PSNR, (2) VMAF vs. decoded source; NR (peach): (3) NIQE, (4) CLIP-IQA, (5) MANIQA. Upscaler rows share 2× delivery resolution, so their NR scores are not comparable to the natively-sized no-upscale row. (T): server-side upscaling seconds per 30-second clip, one representative clip.

Codec KB	Upscaler	(T)↓	(1)↑	(2)↑	(3)↓	(4)↑	(5)↑
DCVC-int16	No upscale	0.0	32.22	73.09	4.56	0.280	0.236
	Bicubic	17.5	30.51	71.21	6.52	0.161	0.206
	S3Diff	1417.4	30.33	66.43	6.19	0.209	0.200
	OSDiff	12536.7	26.73	60.48	3.81	0.599	0.354
	PiD	1204.1	21.13	9.61	4.34	0.358	0.223
H.264 944	No upscale	0.0	29.02	52.82	5.58	0.180	0.188
	Bicubic	17.5	27.83	51.45	7.08	0.131	0.166
	S3Diff	1417.7	28.15	48.72	6.28	0.216	0.187
	OSDiff	13115.7	25.80	50.58	4.57	0.529	0.326
	PiD	1230.3	21.02	8.68	5.00	0.277	0.186
H.265 1077	No upscale	0.0	29.34	57.68	4.62	0.204	0.220
	Bicubic	17.5	28.11	56.19	6.47	0.159	0.204
	S3Diff	1417.4	28.17	53.37	5.91	0.228	0.209
	OSDiff	12536.7	25.73	52.81	3.78	0.590	0.352
	PiD	1204.1	20.92	8.91	4.56	0.326	0.215
AV1 1295	No upscale	0.0	32.01	82.25	4.27	0.289	0.247
	Bicubic	17.3	30.03	79.86	6.21	0.177	0.214
	S3Diff	1490.2	29.63	71.19	5.74	0.233	0.213
	OSDiff	13314.6	26.08	63.35	3.54	0.597	0.368
	PiD	1345.2	20.70	9.71	4.16	0.366	0.226

D. Observations

Table II reports the four upscaling backends within a deployable cost budget plus the no-upscale reference; the video-native models the framework supports are excluded, their per-clip latency approaching a full day, and even OSDiff exceeds three hours per clip. First, the no-upscale reference scores highest on FR metrics within each codec while every generative upscaler trades fidelity for synthesized detail (Section VI-B): VMAF is monotone bicubic→S3Diff→OSDiff→PiD for three of four codecs, and PiD collapses everywhere. On the neural backend, where the grid is complete over all 20 clips, the ordering is unchanged (mean VMAF 70.7 decode, 69.2 bicubic, 65.1 S3Diff, 61.2 OSDiff, 13.3 PiD). Second, at the no-upscale reference the codecs separate into a clear rate-fidelity ordering under this configuration: DCVC-int16 gives the smallest archive (737 KB vs 944–1295 KB) at the highest PSNR (32.22), AV1 leads VMAF (82.3 vs 73.1) at a 76% larger archive, so DCVC-int16 holds the PSNR lead at 57% of AV1’s bitrate, an ordering that holds under every upscaler.

VI. DISCUSSION

A. Why a Dual-Stream Split, Given the Rate Results

The split does not win on rate (Section V-B); its case is structural. It is codec-agnostic: masking and frame selection live outside the encoder, decisive for DCVC, which has a single global quality parameter and no addroi-style mechanism. The two streams also carry independent frame-selection schedules, subsampling the background far more aggressively than the subject, a temporal split a single atomic stream cannot express (modest rate on its own, Section V-B). Independent streams could further support unequal error protection on lossy transport, stated purely as motivation since the current ZIP

container does not packetize them independently. A residual cost falls on the neural path, whose unmasked ROI stream re-encodes out-of-box background; cropping it to the box is future work.

B. Why the Quality Metrics Misrank the Generative Upscalers

Full-reference metrics fail to capture perceived quality once a generative upscaler is in the loop, a case of the perception-distortion tradeoff [24]: the no-upscale reference scores highest on PSNR and VMAF within every codec, yet on visual inspection the generative upscalers produce visibly sharper subjects, since both metrics charge invented texture as error. Scoring the same cells with no-reference metrics (NIQE, CLIP-IQA, MANIQA) inverts the ranking: OSEDIff wins all three in every codec block while bicubic, the most faithful upscaler on VMAF, sits at or near the bottom. The inversion is not simply “more generative is better”: it peaks at OSEDIff rather than the more aggressive PiD, which collapses on the FR side and trails OSEDIff on all three NR metrics. The disagreement is a property of the corpus, not of one clip: on the neural backend, complete over all 20 clips, OSEDIff improves every NR metric over bicubic (NIQE 7.49→4.47, CLIP-IQA 0.190→0.590, MANIQA 0.249→0.370) at a 5.5 dB PSNR cost.

FR metrics remain valid for fidelity-preserving choices, the codec and the bicubic baseline (Table I); generative upscalers should instead be read on the NR columns of Table II. Because a generative upscaler synthesizes plausible rather than measured detail, it can invent markings a viewer reads as a diagnostic cue when no such evidence was captured; we therefore treat upscaled output as a perceptual aid, not photometric evidence, and keep the upscaler a separable, optional module. A human mean-opinion-score study remains the gap.

VII. CONCLUSION

We presented FaunaCodec, a subject-focused compression framework for remote wildlife monitoring: the ROI-aware split holds the subject at high quality while spending far fewer bits on the background, with DCVC-int16 giving the best rate-fidelity operating point in the end-to-end configuration. The split’s value is structural: a codec-agnostic route to region-adaptive quality, in space and time, for a neural backend without a native ROI mode; unequal error protection remains motivation, not demonstrated.

Because the framework withholds bits outside the detected region, it is bounded by the detector’s recall: an undetected subject is compressed as background and cannot be recovered, degrading under fast motion, low nocturnal contrast, and dense multi-subject scenes. We do not benchmark alternative detectors or characterize false-negative rates: detector accuracy is an orthogonal research line, and we build on MegaDetector V6, the standard detector for camera-trap imagery [13]. Future work includes video-native upscalers (Section V-D), human evaluation of generative super-resolution (Section VI-B), cropping the ROI stream to its box (Section VI-A), and evaluating the optional diffusion-based restoration stage the implementation provides.

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